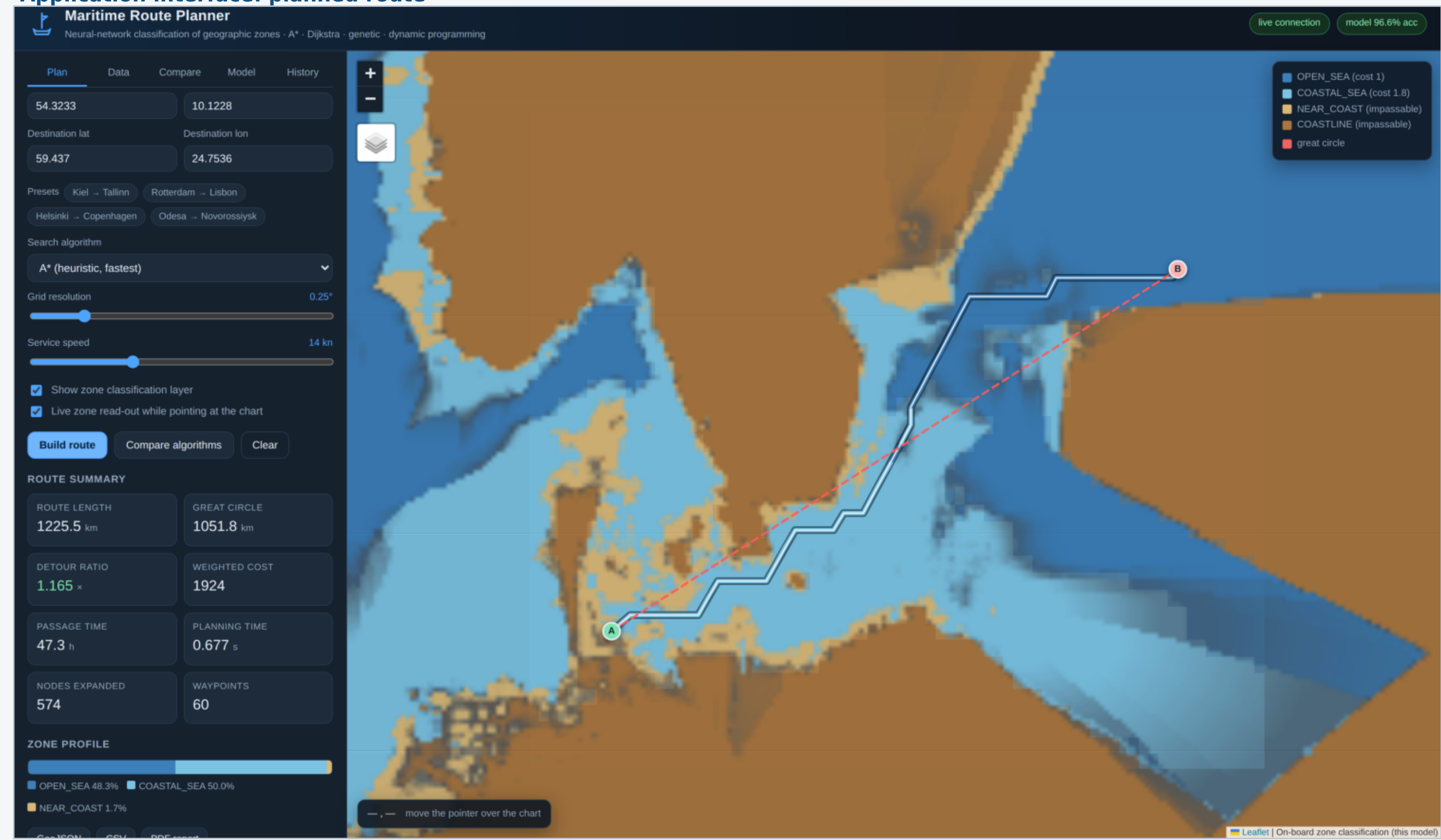
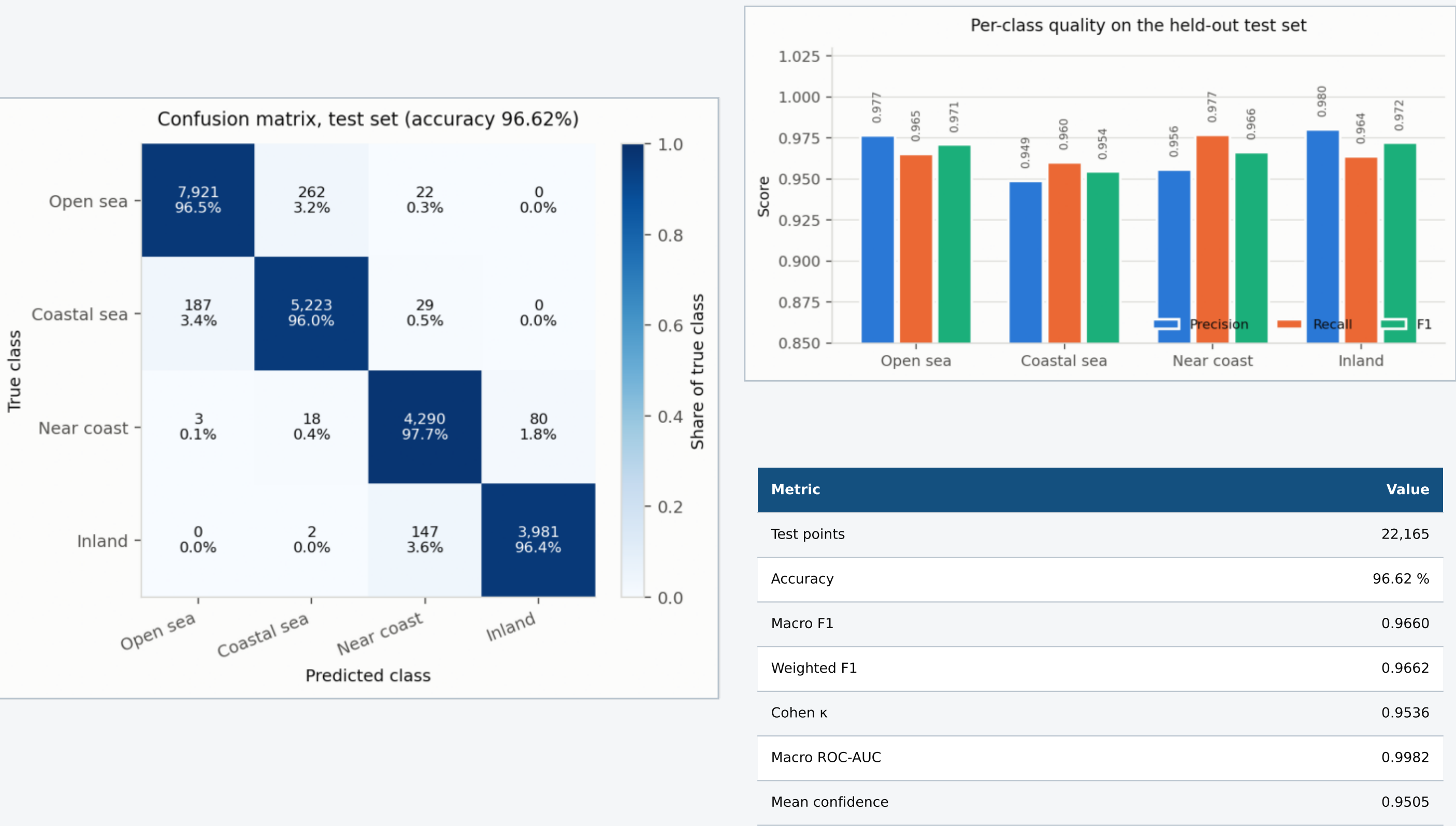


Application interface: planned route

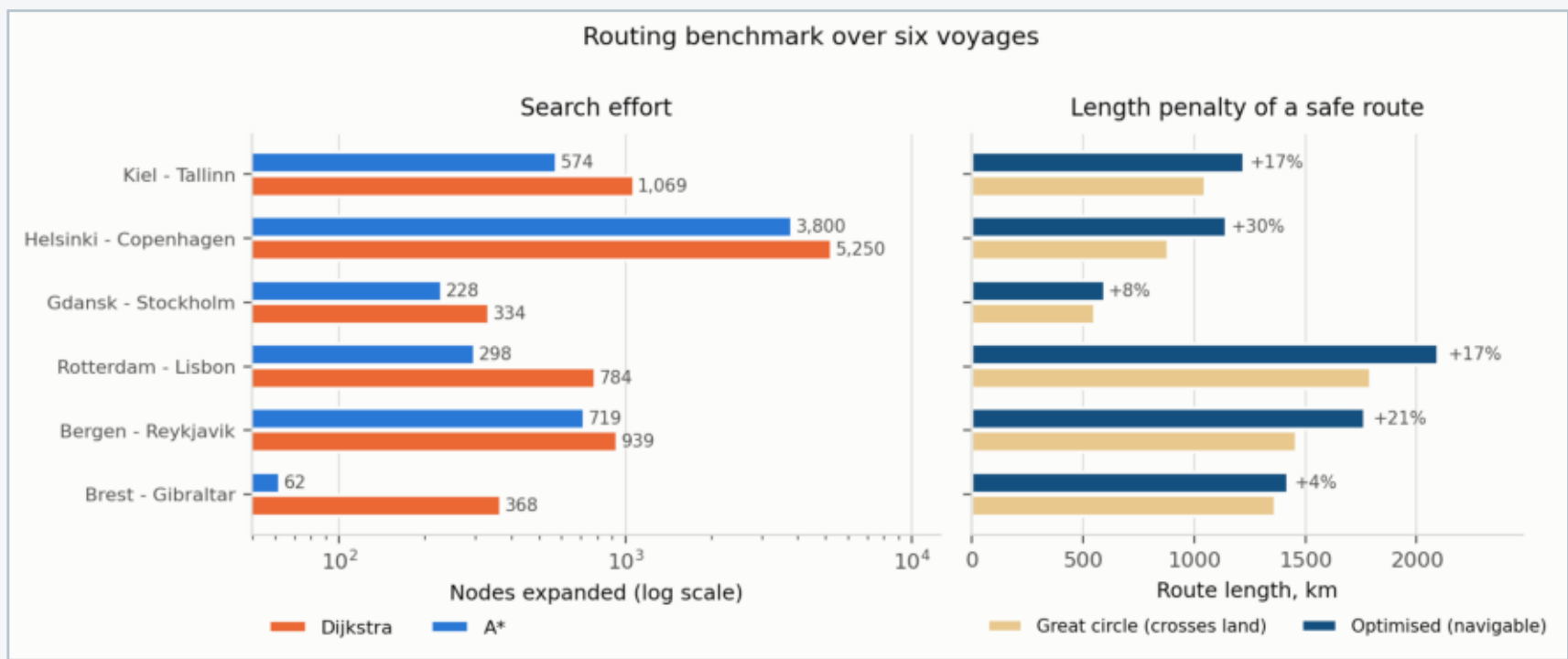


Kiel → Tallinn, A*, lattice 0.25°: the route stays in navigable zones while the great circle (red) crosses Denmark and Sweden

Classification quality on the test set



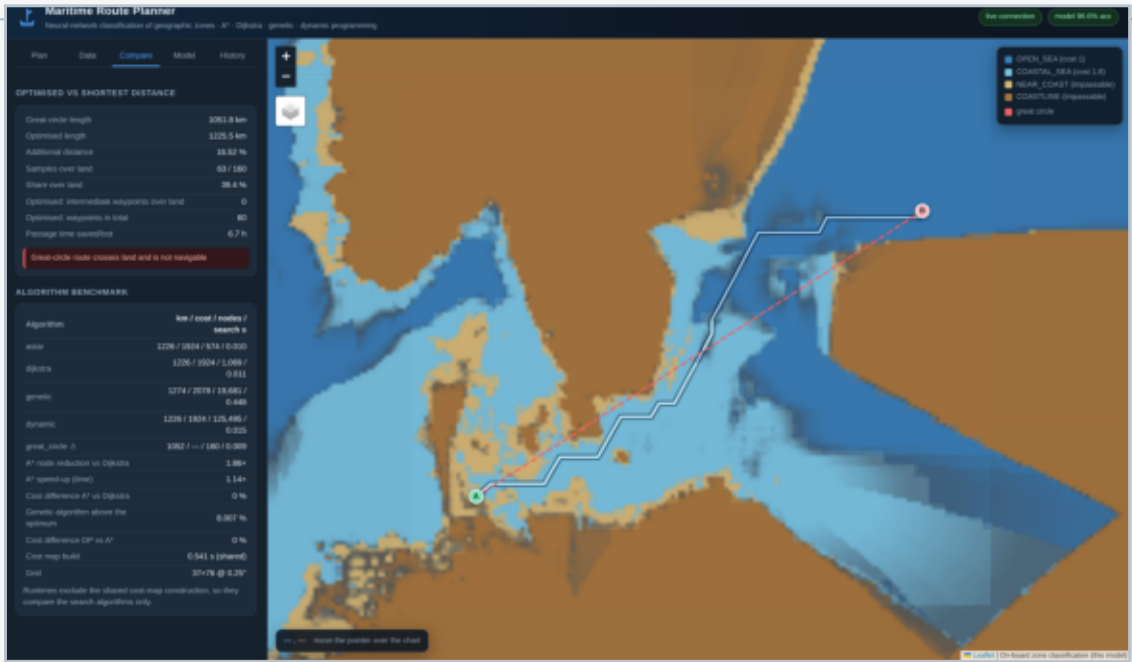
Routing benchmark



Voyage	A* km	Great circle km	Extra	A* nodes	Dijkstra nodes	Search s
Kiel - Tallinn	1,226	1,052	+16.5 %	574	1,069	0.009
Helsinki - Copenhagen	1,146	883	+29.8 %	3,800	5,250	0.028
Gdansk - Stockholm	600	555	+8.1 %	228	334	0.006
Rotterdam - Lisbon	2,098	1,794	+16.9 %	298	784	0.007
Bergen - Reykjavik	1,767	1,458	+21.2 %	719	939	0.010
Brest - Gibraltar	1,423	1,364	+4.3 %	62	368	0.005

Across 6 voyages A* expands 2.43× fewer nodes than Dijkstra and runs 1.14× faster, while both return the identical optimal cost (max deviation 0.0 %). All 6 great-circle routes cross land; the optimised routes are 16.1 % longer on average and fully navigable.

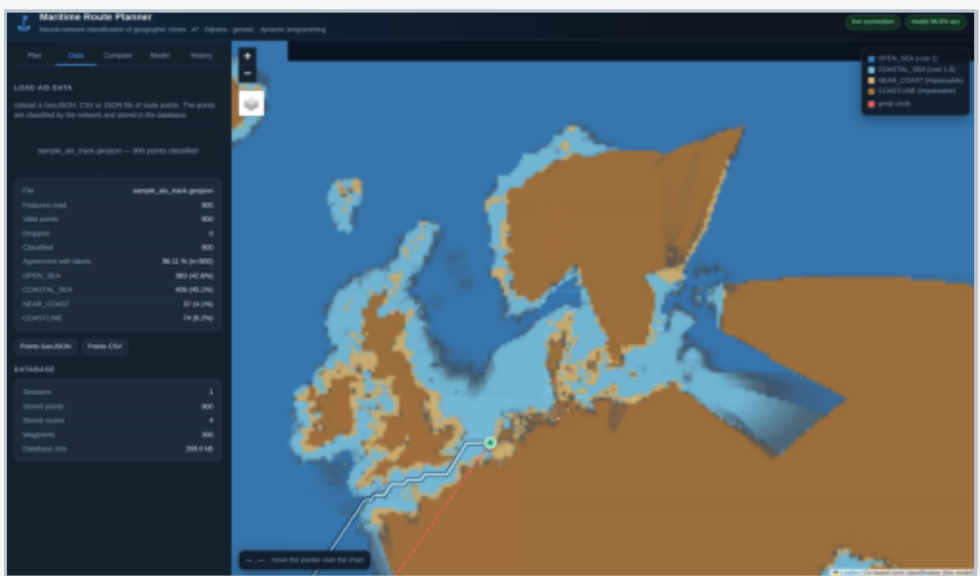
Comparison, uploaded data and stored routes



Algorithm comparison on one shared cost map



Rotterdam → Lisbon, lattice 0.25°



AIS upload: 900 points classified against their labels



Model panel: metrics, matrix, training curves